P4.47

$$V_z = V_{z0} + I_z r_z$$
$$3 = V_{z0} + 5m.100$$
$$V_{z0} = 2.5V$$

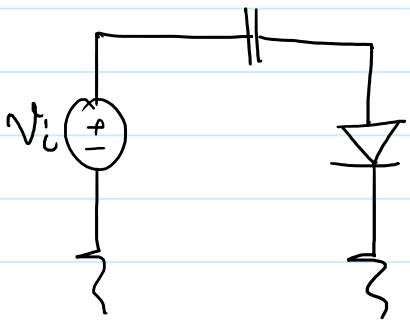


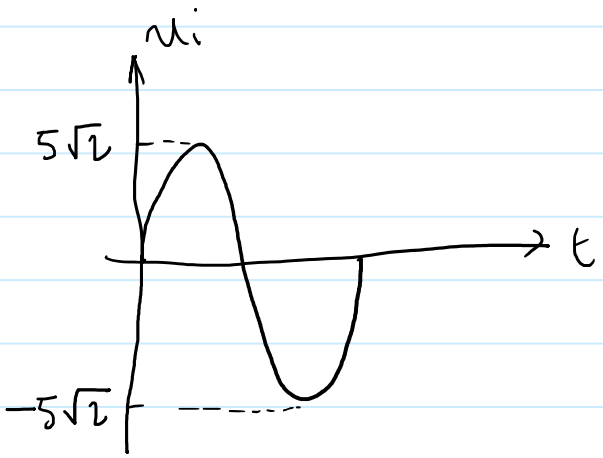
$$P_{max} = 40mW = V_z . I_z = (2.5 + I_z . 100) . I_z$$

$$I_z \cong 11mA \Rightarrow V_z = 3.6V$$

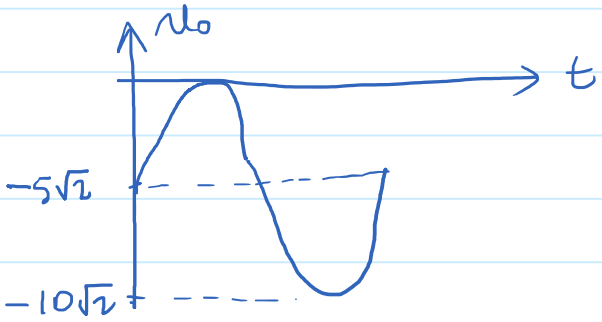
$$I_z = \frac{V_s - V_z}{1k} \Rightarrow V_s = I_z . 1k + V_z$$
$$V_s = 11 + 3.6 = 14.6V$$

P4.86



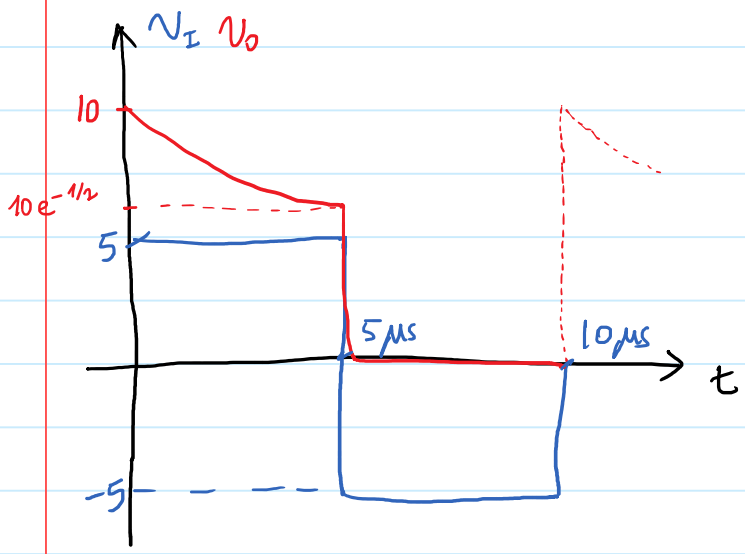


5 rms $\longrightarrow$ $5\sqrt{2}$ peak value



Average $(v_o) = -5\sqrt{2}\,V$

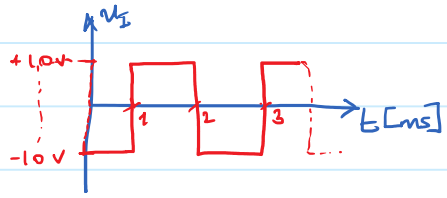
P4.87



$f = 100kHz \rightarrow T = 10\mu s$

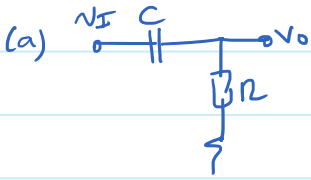
$\tau = RC = 10\mu s$

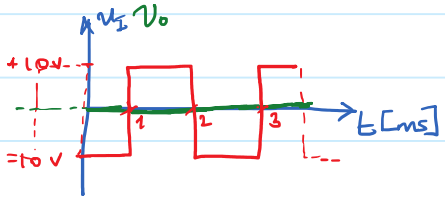
P4.88 - 1



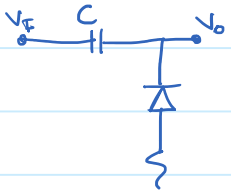
$RC = \tau \gg T = 2ms$

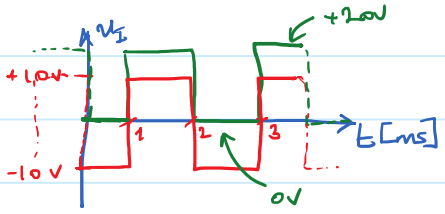
(a)



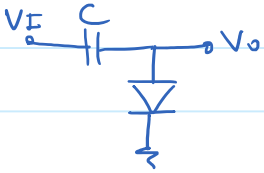


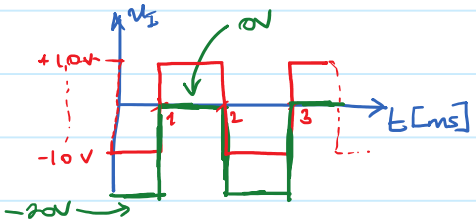
(b)



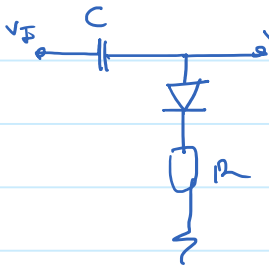


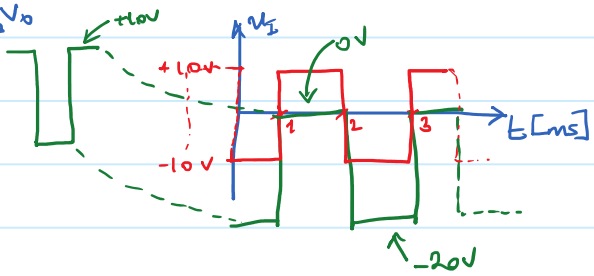
(c)



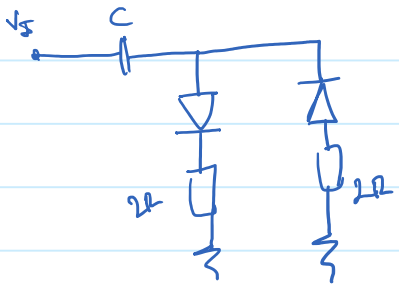


(d)



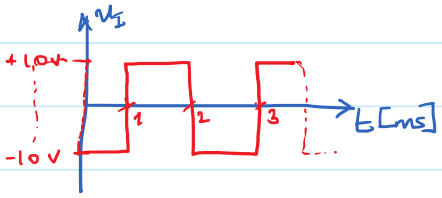


(e)



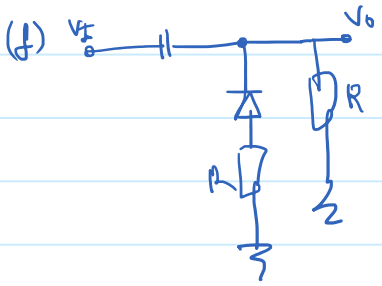
Similar to (a)

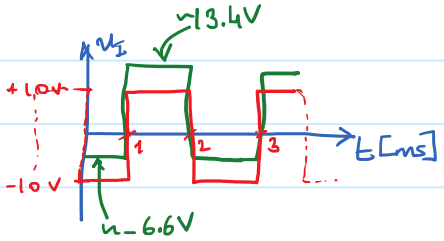
P4.88 - 2



$RC = \tau \gg T = 2ms$

(f)





$$\frac{|V_{omax}|}{R} = \frac{|V_{omin}|}{R} \times 2$$

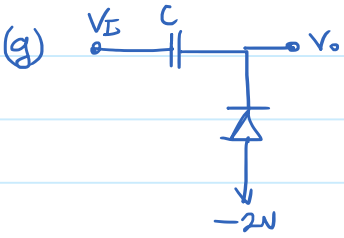
$$|V_{omin}| \times 2 = |V_{omax}|$$

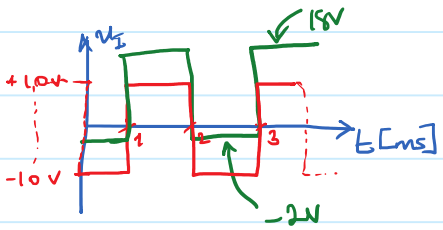
$$|V_{omin}| + |V_{omax}| = 20$$

$$|V_{omin}| \simeq -6.6V$$

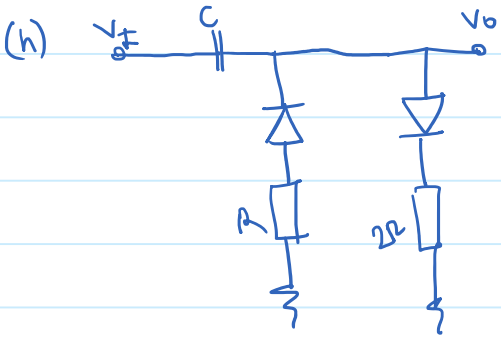
$$|V_{omax}| \simeq 13.4V$$

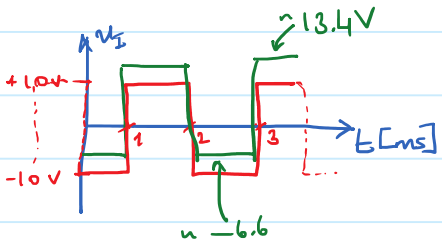
(g)





(h)





$$\frac{|V_{omax}|}{2R} \simeq \frac{|V_{omin}|}{R}$$

$$|V_{omax}| + |V_{omin}| = 20\,V$$